Lab Five-Six – Data Wrangling in R

Tips

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Use the pipe operator (`|>`). It makes the data wrangling tasks much easier to write.
- Use `head(...)` or `glimpse(...)` often to ensure your code did what you wanted it to and that you haven't accidentally deleted their entire dataset with a bad filter.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!

Timeline

- Week of 2/16: Aim to complete the data wrangling sections
- Week of 2/23: Aim to complete the data summaries sections

1 Biomedical Science Data

Kitsberg et al. (2025) collect data on the Human Cytomegalovirus (HCMV). HCMV is a member of the herpesvirus family – the same family that includes: chickenpox, mononucleosis (mono), and cold sores. Prevalence is high with a worldwide seroprevalence of 24% to 100% (Gupta and Shorman 2025). The virus persists in a dormant state for the entire life of those infected. Primary cytomegalovirus infection is often asymptomatic but can present with flu-like symptoms. The infection can reactivate due to stress, aging, or a weakened immune system.

The researchers obtained blood samples from healthy donors, males and females, aged 25-45. They infected the cells of primary (human donor's blood) or THP1 (purchased cell lines) monocytes ($n = 87$ and $n = 112$, respectively) and macrophages ($n = 93$ and $n = 170$, respectively) and counted the number of viral genomes in the nuclei.

Note that the original donor of the THP-1 cell line was a 1-year-old Japanese boy. He was treated for Acute Monocytic Leukemia at Tohoku Hospital Pediatrics (THP-1; the first cell line there). Because they are malignant cancer cells, they can divide indefinitely, breaking the usual cell managment that usually reigns in rogue cells. Therefore, these cells can divide indefinitely in a lab environment. THP-1 cells, and those like them, serve as such a reliable, uniform model for medical research and improve reproducability.

Preliminary Question: Are THP-1 cells a good stand-in for human cells?

Research Question: Is the virus more prevalent in the nucleus of monocytes or macrophages, explaining why HCMV stays dormant in monocytes but goes active in macrophages?

1.1 Data Wrangling

1.1.1 Part A

Go to the website and download the source data https://www.nature.com/articles/s41467-025-68063-y#Sec30.

Describe the process – do not leave out any steps – for how you can load the data for Figure 3b from the manuscript into `R`.

Hint: You can use packages like `readxl` (Wickham and Bryan 2025) but you will find it substantially easier to open the `.xls` file using your computer's spreadsheet software and create a new `.csv` file with the data you need.

Solution

The first thing I did was to open the website and scroll down to source data, then I saved the data and uploaded it to apple numbers. I then converted it to a csv by exporting it to a csv. Next, I deleted the columns with no information and combined the sample columns and number of viruses columns so that both columns had an equal number of entries. Then I added it to my repository and used read.csv() to make sure it uploaded correctly.

```
dat.kitsberg <- read.csv("Fig.3B-Table 1.csv")
```

1.1.2 Part B

The data need to be cleaned. Note that the sample type (Healthy Donor or THP-1), cell type (Monocytes or Macrophages), and treatment (12 Hours Post-Infection or Uninfected) are all recorded in one column `sample`, separated with underscores. Use `separate(...)` from the `tidyverse` package to separate the `sample` column into the `SampleType`, `CellType`, `Treatment` columns. Save the object as `dat.kitsberg.cleaned`.

Hint: Use `?separate(...)` to see how it works.

Solution

```
dat.kitsberg.cleaned = dat.kitsberg |> separate("sample", c("SampleType", "CellType", "Treatment"))
# this passes the original data set through the seperate code adn splits it into different columns
```

1.1.3 Part C

The researchers are only interested in observations in the treatment group (12hpi). Remove uninfected samples from the data. Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `filter(...)` function.

Solution

```
dat.kitsberg.cleaned=dat.kitsberg.cleaned |> filter(Treatment == "12hpi")
# passes the data set through the filter in which treatment must be 12hpi
```

1.1.4 Part D

When `CellType` is "MDM", it is a Monocyte-Derived Macrophages. That is, the cell type should be macrophages which is coded as "mac". Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

Solution

```
dat.kitsberg.cleaned=dat.kitsberg.cleaned |> mutate(
                              if_else(CellType == "MDM", "mac",CellType))
#within the data set, mutate each MDM Celltype "mac," otherwise leave it the same
```

1.2 Part E

The researchers are only interested in monocytes and macrophages. Remove any other cell types. Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `filter(...)` function.

Solution

```
dat.kitsberg.cleaned=dat.kitsberg.cleaned |> filter(CellType %in% c("mac", "mono"))
#pass the data through the filter
# filter is where only "mac" and "mono" are the only possible categories in the CellType column
```

1.3 Part F

Clean up the data entry. Replace the shorthand the researchers used with the neat sample type (Healthy Donor or THP-1), cell type (Monocytes or Macrophages), and treatment (12 Hours Post-Infection or Uninfected). Update

`dat.kitsberg.cleaned` with the result.

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

Solution

```
dat.kitsberg.cleaned=dat.kitsberg.cleaned |>
  mutate(CellType = if_else(CellType == "mono", "Monocytes", "Macrophages")) |>
  mutate(SampleType = if_else(SampleType == "primary", "Healthy Donor", "THP-1"))|>
  mutate(Treatment = if_else(Treatment == "uninfected", "Uninfected", "12 Hours Post-Infection"))
#updates the data set with more precise names with mutate
#I used if_else becuase there were only two categories in each column
```

1.4 Part G

In an `R` code block, combine your answers into one code block that completes the full cleaning process. Save the object as `dat.kitsberg.cleaned`.

Note: You will use this in the next section

Solution

```
dat.kitsberg <- read.csv("Fig.3B-Table 1.csv")
dat.kitsberg.cleaned = dat.kitsberg |>
  separate("sample",c("SampleType", "CellType", "Treatment"))|>
  filter(Treatment == "12hpi")|>
  mutate(CellType =
    if_else(CellType == "MDM", "mac",CellType))|>
  filter(CellType %in% c("mac", "mono"))|>
  mutate(CellType =
    if_else(CellType == "mono", "Monocytes", "Macrophages")) |>
  mutate(SampleType =
    if_else(SampleType == "primary", "Healthy Donor", "THP-1"))|>
  mutate(Treatment =
    if_else(Treatment == "uninfected", "Uninfected", "12 Hours Post-Infection"))
```

1.5 Data Summaries

1.5.1 Part A

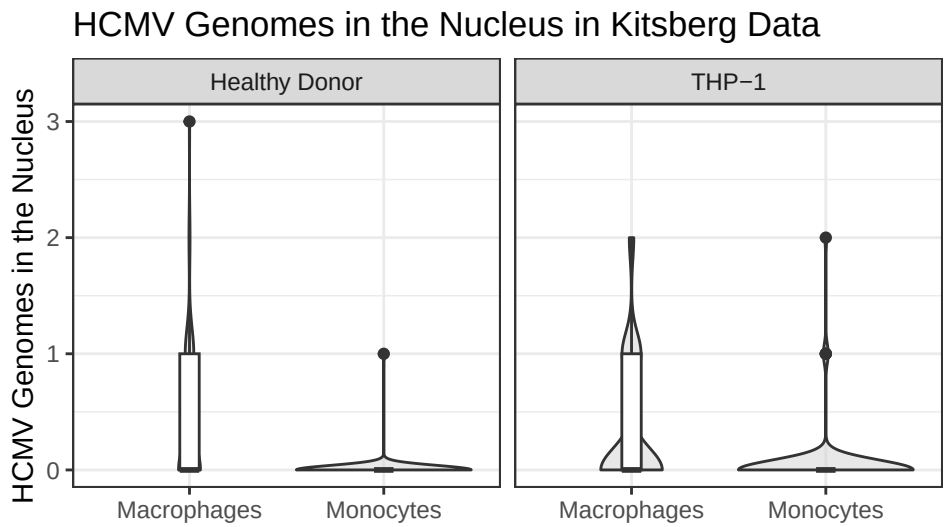
Recreate Figure 3b as depicted in the paper (see https://www.nature.com/articles/s41467-025-68063-y#Fig3). You can ignore the significance comparison because we haven't done that yet.

Hint: I would use `geom_violin(...)` for Figure 3b I also expect your plots to look nicer – better labels, better scaling, etc. Don't forget to use the proper formatting using the syntax in the Quarto cheatsheet.

Solution

```
ggplot(dat.kitsberg.cleaned) + # Specify Data
  geom_violin(aes(x = CellType, # Specify no grouping
                  y = viruses.within.nucleus), # Specify y-axis
                  fill = "lightgrey", alpha=0.5) + # A transparent light grey
  geom_boxplot(aes(x = CellType, # Specify no grouping
                   y = viruses.within.nucleus), # Specify y-axis
                   width = 0.1)+
 theme_bw() + # A cleaner theme
 labs(x = "", # Specify axis labels
      y = "HCMV Genomes in the Nucleus") +
 ggtitle("HCMV Genomes in the Nucleus in Kitsberg Data") + # Plot title
 facet_wrap(~ SampleType, , ncol = 2)
```

Figure 1: Viruses within Nucleus of Cells by Sample Type



This graphical summary demonstrates that in both healthy donors and those with THP-1 there are typically more HCMV genomes in the nucleaus of macrophages. In both the healthy donors and THP-1 donors, there were few instances of the HCMV genomes in the nucleus of the monocytes, both only had a few outliers. However, this graph does not properly represent the data as it does not show the count of the HCMV Genomes in the nucleus of the different cells of different types of patients. A bar plot would be helpful.

1.5.2 Part B

Compute a table containing relevant numerical summaries for Figure 3b.

Hint: I would use `group_by(...)` and `summarize(...)` to complete this. Don't forget to use the proper formatting using `knitr::kable(...)` as described in the Quarto Cheatsheet.

Solution

```
dat.kitsberg.cleaned.summary <- dat.kitsberg.cleaned |>
  group_by(SampleType, CellType) |> #seperates data by sample category
  summarize( #calculates data for a table with the following measures
    n = n(),
    Mean = mean(viruses.within.nucleus, na.rm = T),
    SD = sd(viruses.within.nucleus, na.rm = T),
    Median = median(viruses.within.nucleus, na.rm = T),
    IQR = IQR(viruses.within.nucleus, na.rm = T),
    Skewness = e1071::skewness(viruses.within.nucleus, na.rm = T),
    "Excess Kurtosis" = e1071::kurtosis(viruses.within.nucleus, na.rm = T)
  )
```

Table 1: Viruses within Nucleus of Cells by Sample Type

SampleType	CellType	n	Mean	SD	Median	IQR	Skewness	Excess Kurtosis
Healthy Donor	Macrophages	93	0.366	0.604	0	1	1.690	3.038
Healthy Donor	Monocytes	87	0.011	0.107	0	0	9.008	80.069
THP-1	Macrophages	170	0.329	0.573	0	1	1.530	1.313
THP-1	Monocytes	112	0.045	0.247	0	0	5.968	38.005

This table demonstrates that in patients with THP-1, the average amount of viruses within the nucleus of cells is higher than healthy donors in the monocytes and lower in macrophages. It is important to note that all of these data distributions are right skewed and are leptokurtic, demonstrating that the prescence of viruses within any cells are fairly uncommon, which is further supported by the median of 0 (a more reliable measure of center than the mean with the amount of skew).

2 Ship Valuation

Figure 2: This is a ship-shipping ship shipping shipping ships.

Bal, Akan, and Gencer (2025) aim to build a model for predicting the value of Supramax and Ultramax ships using both current and past market data. Their goal was to build a model that only uses information anyone can easily look up.

The researchers collected 17 years of real ship sale prices (2005–2022), covering $n = 1819$ Supramax and Ultramax ship sales.

Research Question: Can we predict Supramax and Ultramax ship prices?

2.1 Data Wrangling

2.1.1 Part A

I downloaded the data from linked in the article (https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0319073). I have included the data in this lab folder as `ShipValuation-Bal25.csv`. Load the data into `R`.

Solution

```
dat.bal = read.csv("ShipValuation-Bal25.csv")
```

2.1.2 Part B

The price of each Supramax and Ultramax ship is reported on the $\log_{10}$ scale. Create a new column called `price` that contains the price in millions of dollars.

$$\text{Price} = \frac{10^{\text{SPLog}}}{1000000}$$

Save the object as `dat.bal.cleaned`.

Hint: I would use the `mutate(...)` function.

Solution

```
dat.bal.cleaned = dat.bal |> mutate(Price = (10^SPLog)/1000000)
#For every price value in dat.bal, that value passes through that equation
```

2.1.3 Part C

The authors recorded the country that each Supramax or Ultramax ship was built in in the columns `Japan`, `China, People's Republic Of`, `Philippines`, `Korea, South`, `Others Country Build`. These columns are marked `1` to indicate that is the country of origin and `0` otherwise. Create a new column called `OriginCountry` that contains a clean labels for the country in which the ship was built. Update `dat.bal.cleaned` with the result.

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(OriginCountry =
  #create a new column, "OriginCountry" with mutate
    case_when(
      Japan == 1 ~ "Japan",
      China..People.s.Republic.Of == 1 ~ "China",
      Philippines == 1 ~ "Philippines",
      Korea..South == 1 ~ "S. Korea",
      Others.Country.Build == 1 ~ "Other Country"
      )
    )
    #when a given country column is 1, make that country's name the value in the new column
```

2.1.4 Part D

The authors recorded each Supramax and Ultramax ship's main engine manufacturer in the columns `Caterpillar`, `MAN-B&W`, `Wartsila`, `Mitsubishi`. These columns are marked `1` to indicate that is the manufacturer and `0` otherwise. Create a new column called `Manufacturer` that contains a clean labels for the manufacturer. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous question.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |> mutate(
  Manufacturer = case_when(
      Caterpillar == 1 ~ "Caterpillar",
      MAN.B.W == 1 ~ "MAN-B&W",
      Wartsila == 1 ~ "Wartsila",
      Mitsubishi == 1 ~ "Mitsubishi",
      )
    )
    #See Part C for logic behind this
```

2.1.5 Part E

The researchers recorded whether each Supramax and Ultramax ship is currently covered by a Protection and Indemnity (P&I) Club. This can be an indicator that a ship meets the minimum seaworthiness and safety requirements of the insurer.

The authors recorded P&I clubs in the columns `Japan P&I Club`, `Gard`, `West of England`, `North of England`, `UK P&I`, `Britannia P&I`, `London P&I Club`, `Skuld`, `Steamship Mutual`, `Others`, `Unknown`. These columns are marked `1` to indicate that the ship is insured by that P&I club and `0` otherwise. Create a new column called `PI.club` that contains a clean labels for the P&I club. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |> mutate(
  PI.club = case_when(
```

```
      Japan.P.I.Club == 1 ~ "Japan P&I Club",
      Gard == 1 ~ "Gard",
      West.of.England == 1 ~ "W of England",
      North.of.England == 1 ~ "N of England",
      UK.P.I == 1 ~ "UK P&I",
      Britannia.P.I == 1 ~ "Britannia P&I",
      London.P.I.Club == 1 ~ "London P&I",
      Skuld == 1 ~ "Skuld",
      Steamship.Mutual == 1 ~ "Steamship Mutual",
      Others == 1 ~ "Other",
      Unknown == 1 ~ "Unknown"
      )
    )
    #See Part C for logic behind this
```

2.1.6 Part F

The researchers recorded whether each Supramax and Ultramax ship is classed by a society that requires a ship to be safe, structurally sound, and built/maintained to specific engineering standards.

The authors recorded class society in the columns `Nippon Kaiji Kyokai (IACS)`, `Lloyd's Register`, `Bureau Veritas (IACS)`, `Det Norske Veritas (IACS)`, `American Bureau of Shipping (IACS)`, `Others Class`, `With Two Classes`, `Disclassed`. These columns are marked `1` to indicate that the ship is classed by that soceity and `0` otherwise. Create a new column called `Class` that contains a clean labels for the class. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

```
dat.bal.cleaned = dat.bal.cleaned |> mutate(
  Class = case_when(
      Nippon.Kaiji.Kyokai..IACS. == 1 ~ "Nippon Kaiji Kyokai (IACS)",
      Lloyd.s.Register == 1 ~ "Lloyd's Register",
      Bureau.Veritas..IACS. == 1 ~ "Bureau Veritas (IACS)",
      Det.Norske.Veritas..IACS. == 1 ~ "Det Norske Veritas (IACS)",
      American.Bureau.of.Shipping..IACS. == 1 ~ "American Bureau of Shipping (IACS)",
      With.Two.Classes == 1 ~ "Two Classes",
      Disclassed == 1 ~ "Disclassed",
      Others.Class == 1 ~ "Other"
      )
    )
    #See Part C for logic behind this
```

2.1.7 Part G

The researchers recorded the flag state of each Supramax and Ultramax ship. Each flag state enforces its own set of safety, environmental, and crew welfare rules and regulations. Ships under certain flag states may be more or less valuable depending on how much the flag affects the buyer's trust in the condition of the boat.

The authors recorded flag states in the columns `Panama`, `Marshall Islands`, `Liberia`, `Malta`, `Hong Kong`, `China`, `Singapore`, `Others Flag`, `With two flags`. These columns are marked `1` to indicate that the ship has that flag state and `0` otherwise. Create a new column called `FlagState` that contains a clean labels for the flag state. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

```r
dat.bal.cleaned = dat.bal.cleaned |> mutate(
  FlagState = case_when(
      Panama == 1 ~ "Panama",
      Marshall.Islands == 1 ~ "Marshall Islands",
      Liberia == 1 ~ "Liberia",
      Malta == 1 ~ "Malta",
      Hong.Kong..China == 1 ~ "Hong Kong",
      Singapore == 1 ~ "Singapore",
      Others.Flag == 1 ~ "Other",
      With.two.flags == 1 ~ "Two Flags"
      )
    )
    #See Part C for logic behind this
```

2.2 Part H

In an `R` code block, combine your answers into one code block that completes the full cleaning process. Save the object as `dat.bal.cleaned`.

Note: You will use this in the next section

Solution

```r
dat.bal.cleaned = dat.bal |> mutate(Price = (10^SPLog)/1000000)|>
  drop_na() |> #gets rid of empty cells
  mutate(OriginCountry =
   # creates new column 'Origin Country' containing categories:
   case_when(
      Japan == 1 ~ "Japan",
      China..People.s.Republic.Of == 1 ~ "China",
      Philippines == 1 ~ "Philippines",
      Korea..South == 1 ~ "S. Korea",
      Others.Country.Build == 1 ~ "Other Country"
      )
    ) |>
  mutate(Manufacturer =
   # creates new column 'Manufacturer' containing categories:
   case_when(
      Caterpillar == 1 ~ "Caterpillar",
      MAN.B.W == 1 ~ "MAN-B&W",
      Wartsila == 1 ~ "Wartsila",
      Mitsubishi == 1 ~ "Mitsubishi",
      )
    )|>
  mutate(PI.club =
   # creates new column 'PI.club' containing categories:
   case_when(
      Japan.P.I.Club == 1 ~ "Japan P&I Club",
      Gard == 1 ~ "Gard",
      West.of.England == 1 ~ "W of England",
      North.of.England == 1 ~ "N of England",
      UK.P.I == 1 ~ "UK P&I",
      Britannia.P.I == 1 ~ "Britannia P&I",
      London.P.I.Club == 1 ~ "London P&I",
      Skuld == 1 ~ "Skuld",
      Steamship.Mutual == 1 ~ "Steamship Mutual",
      Others == 1 ~ "Other",
      Unknown == 1 ~ "Unknown"
```

```r
      )
    ) |>
  mutate(Class =
   # creates new column 'Class' containing categories:
   case_when(
      Nippon.Kaiji.Kyokai..IACS. == 1 ~ "Nippon Kaiji Kyokai (IACS)",
      Lloyd.s.Register == 1 ~ "Lloyd's Register",
      Bureau.Veritas..IACS. == 1 ~ "Bureau Veritas (IACS)",
      Det.Norske.Veritas..IACS. == 1 ~ "Det Norske Veritas (IACS)",
      American.Bureau.of.Shipping..IACS. == 1 ~ "American Bureau of Shipping (IACS)",
      With.Two.Classes == 1 ~ "Two Classes",
      Disclassed == 1 ~ "Disclassed",
      Others.Class == 1 ~ "Other"
      )
    )|>
  mutate(FlagState =
   # creates new column 'FlagState' containing categories:
   case_when(
      Panama == 1 ~ "Panama",
      Marshall.Islands == 1 ~ "Marshall Islands",
      Liberia == 1 ~ "Liberia",
      Malta == 1 ~ "Malta",
      Hong.Kong..China == 1 ~ "Hong Kong",
      Singapore == 1 ~ "Singapore",
      Others.Flag == 1 ~ "Other",
      With.two.flags == 1 ~ "Two Flags"
      )
    )
```

2.3 Data Summaries

First, I provide a short description of the other variables in their data. Note that the size ordering of types of ships discussed below is:

$$\text{Handysize} < \text{Supramax} < \text{Ultramax} < \text{Panamax} < \text{Capesize}$$

- `DWT`: The deadweight tonnage, which is the maximum weight each Supramax and Ultramax ship can safely carry in metric tons. This gives a measure of how much cargo, fuel, fresh water, ballast water, provisions, and crew a ship can handle.
- `Yas`: The age in years. New Supramax and Ultramax ships were recorded as having age 1.
- `Enginges RPM`: Revolutions per minute of each Supramax and Ultramax ship's main propulsion engine. Generally speaking, lower-RPM engines are more expensive because they are more fuel efficient.
- `BACI`: The Baltic Capesize Index. This describes freight rates for Capesize dry bulk ships, which reflects supply and demand. When BACI is high, Capesize ships are more profitable in shipping, which could positively affect Supramax and Ultramax ship values as demand for dry bulk commodities might be high and large shipments could be split into several Supramax and Ultramax shipments.
  Note: This reports the current BACI. The data also contain the BACI form 1, 2, and 3 months before the sale (`BACI-1`, `BACI-2`, `BACI-3`).
- `BPNI`: The Baltic Panamax Index. This describes the cost to ship dry bulk commodities (e.g., coal or grain) on Panamax class ships. When BPNI is high, we expect Panamax ship prices to go up. Supramax and Ultramax ship values tend to follow this trend as charterers switch to these smaller vessels when Panamax rates become too expensive.
  Note: This reports the current BPNI. The data also contain the BPNI form 1, 2, and 3 months before the sale (`BPNI-1`, `BPNI-2`, `BPNI-3`).
- `BSIS`: The Baltic Supramax Index. This describes freight rates for Supramax vessels, which reflects supply and demand. When BSIS is high, Supramax and Ultramax prices are usually higher because they are more profitable in shipping.

Note: This reports the current BSIS. The data also contain the BSIS form 1, 2, and 3 months before the sale (`BSIS-1`, `BSIS-2`, `BSIS-3`).

- `BHSI`: The Baltic Handysize Index. This describes freight rates for handysize vessels. When BHSI is high, Supramax and Ultramax ship prices are usually higher because the prices move together.
  Note: This reports the current BHSI. The data also contain the BHSI form 1, 2, and 3 months before the sale (`BHSI-1`, `BHSI-2`, `BHSI-3`).
- `LIBOR`: The London InterBank Offered Rate. This describes the average reported interest rate at which banks would lend US dollars for 3 months.

2.3.1 Part A

Choose a categorical variable. Create a plot and corresponding numerical summary to assess whether or how ship prices vary across the levels of the selected categorical variable.

Solution

```r
ggplot(dat.bal.cleaned) + # Specify Data
geom_violin(aes(x = OriginCountry, # Specify no grouping
                y = Price), # Specify y-axis
                fill = "lightgrey", alpha=0.5) + # A transparent light grey
geom_boxplot(aes(x = OriginCountry, # Specify no grouping
                 y = Price), # Specify y-axis
                 width = 0.1)+
theme_bw() + # A cleaner theme
labs(x = "Origin Country", # Specify axis labels
     y = "Price (Millions of USD)")
```

Figure 3: Prices by Origin Country

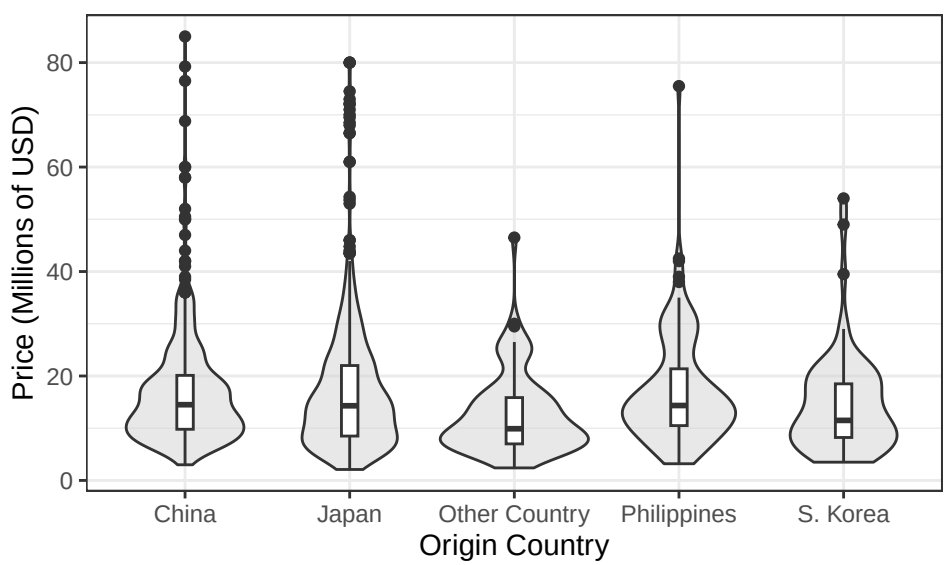


Figure 3 demonstrates that most ship prices are skewed right with many outliers. Generally, most countries' medians are between 10 and 15 million USD. Most of these countries, with maybe the exception of Japan, have interquartile ranges of about 10 million USD. From this graph, there doesn't seem to be major differences in ship prices due to origin country, however, China and Japan have the greatest amount of outliers with multiple ships costing more than $50 million.

```r
OriginCountry.summary <- dat.bal.cleaned |>
  group_by(OriginCountry) |>
  summarize(
    n = n(),
    Missing = sum(is.na(Price)),
    Mean = mean(Price, na.rm = T),
    SD = sd(Price, na.rm = T),
    Median = median(Price, na.rm = T),
    IQR = IQR(Price, na.rm = T),
    Skewness = e1071::skewness(Price, na.rm = T),
    "Excess Kurtosis" = e1071::kurtosis(Price, na.rm = T)
  )
```

Table 2: Prices by Origin Country.

OriginCountry	n	Missing	Mean	SD	Median	IQR	Skewness	Excess Kurtosis
China	608	0	16.824	10.395	14.50	10.315	2.306	8.683
Japan	876	0	16.958	11.859	14.30	13.500	2.091	6.618
Other Country	86	0	12.583	7.546	9.90	8.863	1.596	3.437
Philippines	130	0	17.074	10.256	14.35	10.875	1.891	6.729
S. Korea	66	0	14.639	9.740	11.50	10.275	1.837	4.452

Table 2 provides some greater insight as to why China and Japan may have many more outliers: they produce significantly more ships. The massive jump in number of ships produced between the Phillippines and China could explain the massive jump in number of outliers. This greater number of outliers can also explain why the means of China and Japan are closer to $17 million while the others are below $15 million (with the exception of the Philippines which just has one major outlier pulling the mean). Because the skewness of all of these distributions are fairly high, I believe the median and IQR are more accurate measures of this data. This distribution is also leptokurtic (the excess kurtosis of all countries is positive), further supporting the need to use median and interquartile range. Unlike with the violin plot, it is much more clear that China, Japan, and the Philippines are typically more expensive with a medians over $14 million, while South Korea ship prices have a median of $11.5 million and other countries have a median of $9.9 million. Based on the IQR, Japan has the greatest amount of variation, as seen in the violin plot, other countries has the least variation, and China, the Philippines, and South Korea all have around the same variation with interquartile ranges between $10 and $11 Million. Therefore, ship prices could differ based on country of origin with China, Japan, and the Phillippines typically being the most expensive. However, before any majors conclusions can be drawn, it is worth noting that these ship price variations could be caused by different countries producing different types of ships or some other confounding variable.

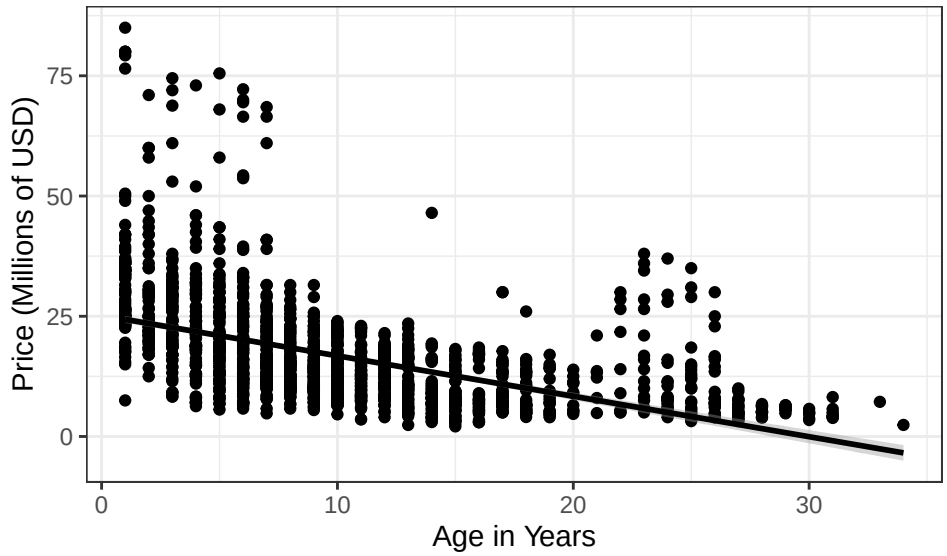
2.3.2 Part B

Choose a quantitative variable. Create a plot and corresponding numerical summary to assess whether or how ship prices vary with the selected quantitative variable.

Solution

```r
ggplot(dat.bal.cleaned) + # Specify Data
geom_point(aes(x = Yas, # Specify x-axis
               y = Price)) + # Specify y-axis
geom_smooth(aes(x = Yas, # Specify x-axis
                y = Price), # Specify y-axis
                method = "lm", # Specify best fit line
                color = "black") + # Specify line color
theme_bw() + # A cleaner theme
labs(x = "Age in Years", # Specify axis labels
     y = "Price (Millions of USD)")
```

Figure 4: Years' impact on Price



In Figure 4, the line has a very flat negative slope, showing a possible negative relationship between price and the age of the ships. There is greater variation in price within the first ten years after a ship is manufactured, which could be caused by Supramax and Ultramax being recorded as having age 1. But there is also greater variation around the 25 year mark, so there could be another factor aside from just ship type. But generally, as age increases, the price of ships decrease.

```
Age.summary = dat.bal.cleaned |>
summarize(
  Pearson = cor(Price, Yas, method="pearson"),
  Spearman = cor(Price, Yas, method="spearman"),
  Kendall = cor(Price, Yas, method="kendall")
        )
```

Table 3: Price (Millions of $) by Age (years).

Pearson	Spearman	Kendall
-0.513	-0.655	-0.49

Table 3 shows that there is a moderately negative relationship between age and price. It is likely monotonic rather than linear as the Pearson correlation is not as strong as the Spearman correlation. This is supported as the Kendall Tau is also shows a negative monotonic correlation

3 Citations

I used the patchwork package (Pedersen 2025) to help compose plots, the tidyverse package (Wickham et al. 2019) to help create tables and graphs, the knitr package (Xie 2025) for cleanly organized graphs and tables, the e1071 package (Meyer et al. 2025) to find excess kurtosis and skewness, all while using R (R Core Team 2025) as my program.

References

Bal, Elif Tuçe, Ercan Akan, and Huseyin Gencer. 2025. "A Novel Approach to Ship Valuation Prediction: An Application to the Supramax and Ultramax Secondhand Markets." *PLoS One* 20 (5): e0319073.

Gupta, Mohit, and Mahmoud Shorman. 2025. "Cytomegalovirus Infections." In *StatPearls [Internet]*. StatPearls Publishing.

Kitsberg, Yaarit, Aharon Nachshon, Tamar Arazi, Karin Broennimann, Tal Fisher, Alexander Wainstein, Yaara Finkel, Noam Stern-Ginossar, and Michal Schwartz. 2025. "Viral Entry Shapes HCMV Latency Establishment." *Nature Communications*.

Meyer, David, Evgenia Dimitriadou, Kurt Hornik, Andreas Weingessel, and Friedrich Leisch. 2025. *E1071: Misc Functions of the Department of Statistics, Probability Theory Group (Formerly: E1071), TU Wien*. https://doi.org/10.32614/CRAN.package.e1071.

Pedersen, Thomas Lin. 2025. *Patchwork: The Composer of Plots*. https://doi.org/10.32614/CRAN.package.patchwork.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. "Welcome to the tidyverse." *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.

Wickham, Hadley, and Jennifer Bryan. 2025. *Readxl: Read Excel Files*. https://doi.org/10.32614/CRAN.package.readxl.

Xie, Yihui. 2025. *Knitr: A General-Purpose Package for Dynamic Report Generation in R*. https://yihui.org/knitr/.